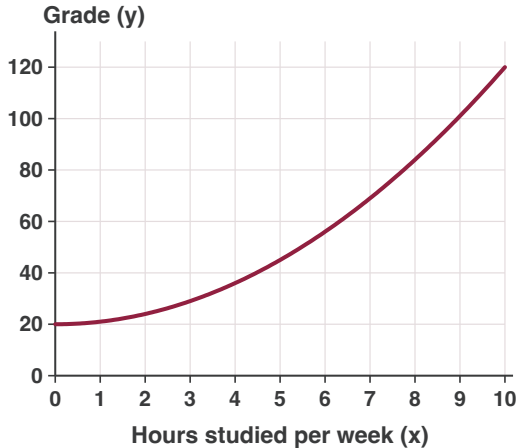


Limits and the Derivative

ECON 441: Introduction to Mathematical Economics
Lecture 8

Div Bhagia

(Hypothetical) Production Function for Grades



$$y = f(x) = x^2 + 20$$

How much can your grade increase if you study for one additional hour per week?

Depends on how much you are studying right now!

Average Rate of Change

Use Δ to denote change:

$$\Delta x = x_1 - x_0$$

Change in y per unit change in x :

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

Average Rate of Change

$$y = f(x) = x^2 + 20$$

What happens if you go from 6 to 8 hours of studying?

$$x_0 = 6, x_1 = 8 \rightarrow \Delta x = x_1 - x_0 = 2$$

Total change in grades:

$$f(x_0 + \Delta x) - f(x_0) =$$

Per hour change in grade:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} =$$

The Derivative

Usually we are interested in minuscule changes from x_0 .

The **derivative** of a function is defined as:

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

Note that

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

The Derivative

For the function $y = f(x) = x^2 + 20$:

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \\ &= \frac{(x_0 + \Delta x)^2 + 20 - (x_0^2 + 20)}{\Delta x} \\ &= \frac{x_0^2 + (\Delta x)^2 + 2x_0\Delta x - x_0^2}{\Delta x} \\ &= 2x_0 + \Delta x\end{aligned}$$

Then the derivative is

$$\frac{dy}{dx} = f'(x_0) = \lim_{\Delta x \rightarrow 0} (2x_0 + \Delta x) = 2x_0$$

The Derivative

The derivative:

$$\frac{dy}{dx} = f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

Alternatively,

$$\frac{dy}{dx} = f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

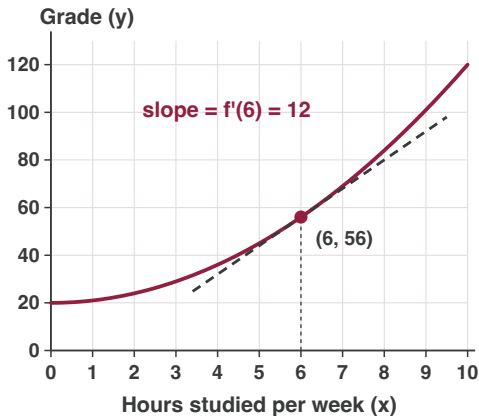
The Derivative

For the function $y = f(x) = x^2 + 20$:

$$\begin{aligned}\frac{dy}{dx} &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{x \rightarrow x_0} \frac{x^2 + 20 - x_0^2 - 20}{x - x_0} \\ &= \lim_{x \rightarrow x_0} \frac{x^2 - x_0^2}{x - x_0} \\ &= \lim_{x \rightarrow x_0} (x + x_0) = 2x_0\end{aligned}$$

Derivative = Slope of the Tangent Line

$$y = f(x) = x^2 + 20 \rightarrow f'(x) = 2x$$



Concept of a Limit

We say that L is the **limit** of $f(x)$ at a , i.e.

$$\lim_{x \rightarrow a} f(x) = L$$

if $f(x)$ approaches L as x approaches a from any direction.

Note that we don't actually set $x = a$.

Also, for the limit to exist at a point we need the function to approach the same value from both directions.

Concept of a Limit

Left-side limit: if x approaches a from the left side

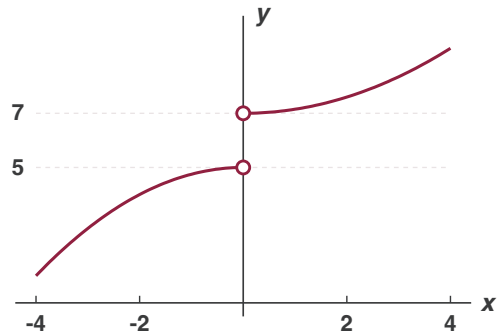
$$\lim_{x \rightarrow a^-} f(x)$$

Right-side limit: if x approaches a from the right side

$$\lim_{x \rightarrow a^+} f(x)$$

Only when both the left-side and right-side limits have a common finite value do we say that the limit exists.

Example: Limit Doesn't Exist



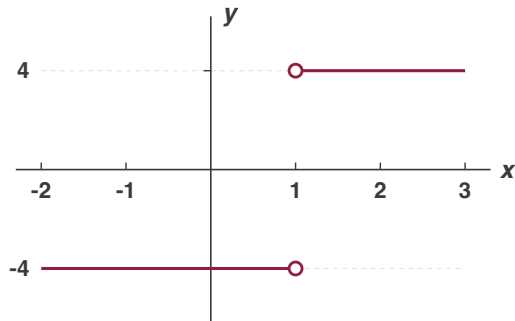
Limit: Another Example

$$f(x) = \frac{4x - 4}{|x - 1|}$$

$$\lim_{x \rightarrow 1^-} f(x) =$$

$$\lim_{x \rightarrow 1^+} f(x) =$$

Limit: Another Example



Homework Questions

- Exercise 6.2: 2, 3

Textbook reference: 6.2-6.4